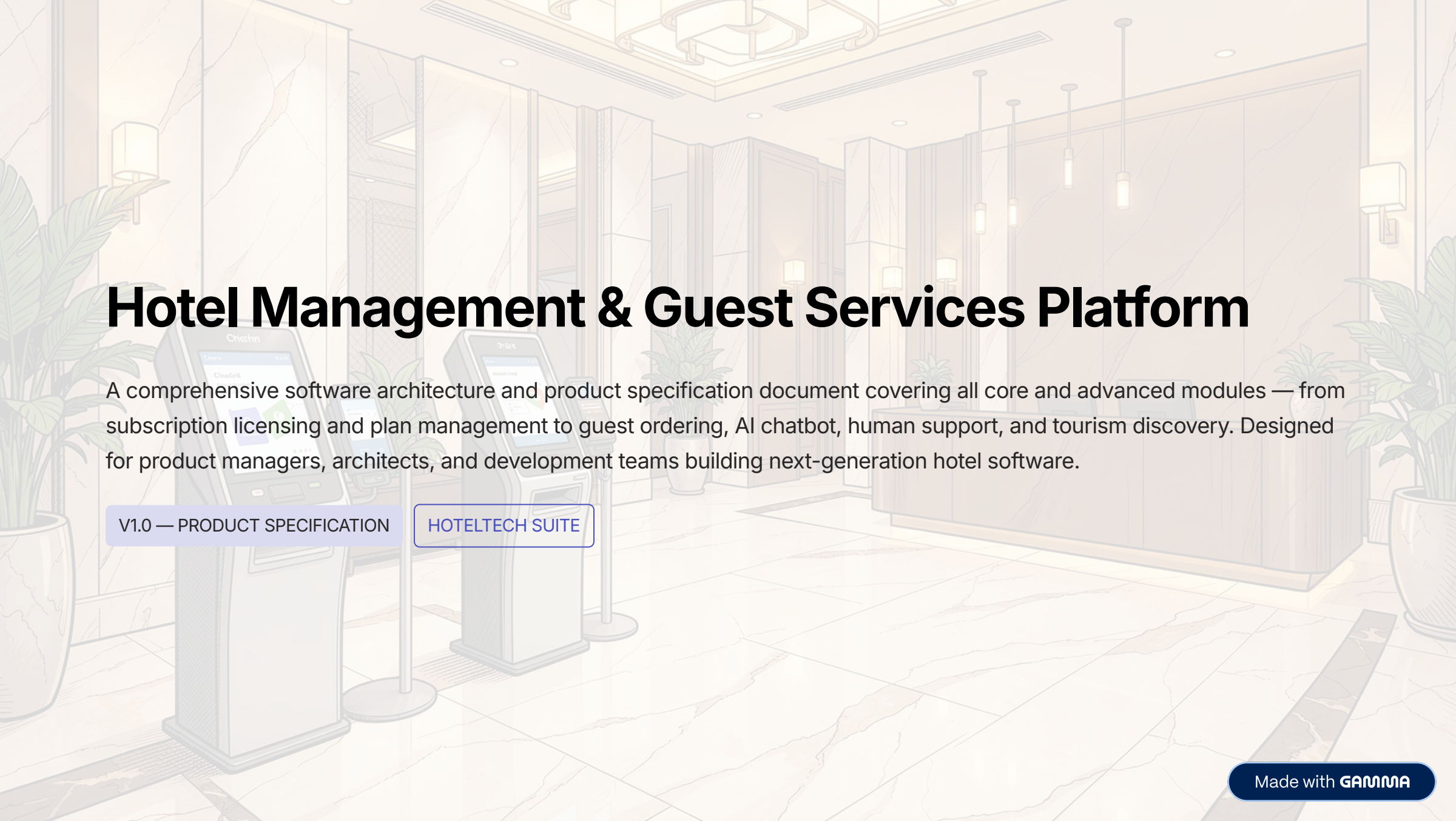




Hotel Management & Guest Services Platform

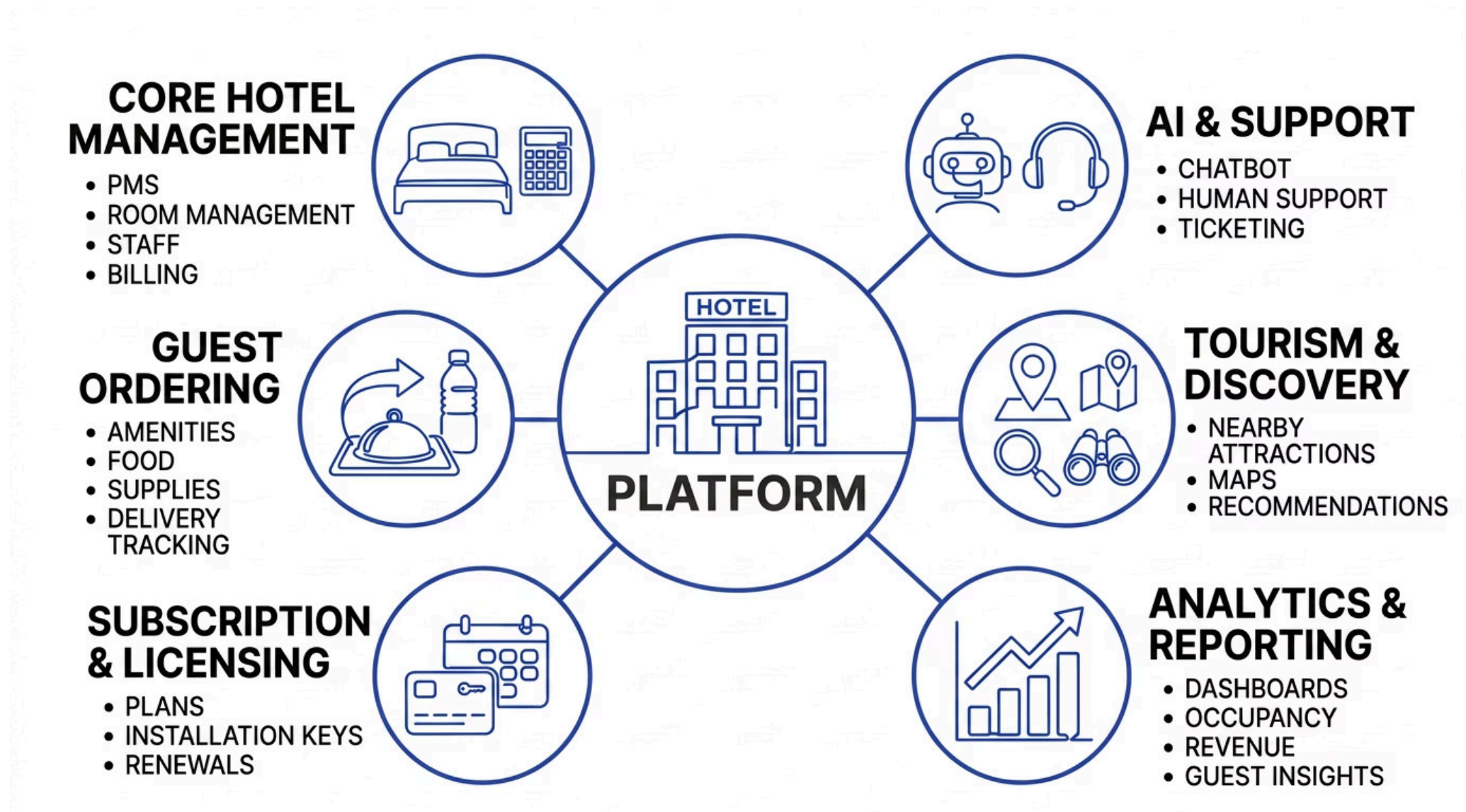
A comprehensive software architecture and product specification document covering all core and advanced modules — from subscription licensing and plan management to guest ordering, AI chatbot, human support, and tourism discovery. Designed for product managers, architects, and development teams building next-generation hotel software.

V1.0 — PRODUCT SPECIFICATION

[HOTELTECH SUITE](#)

Platform Overview & Module Map

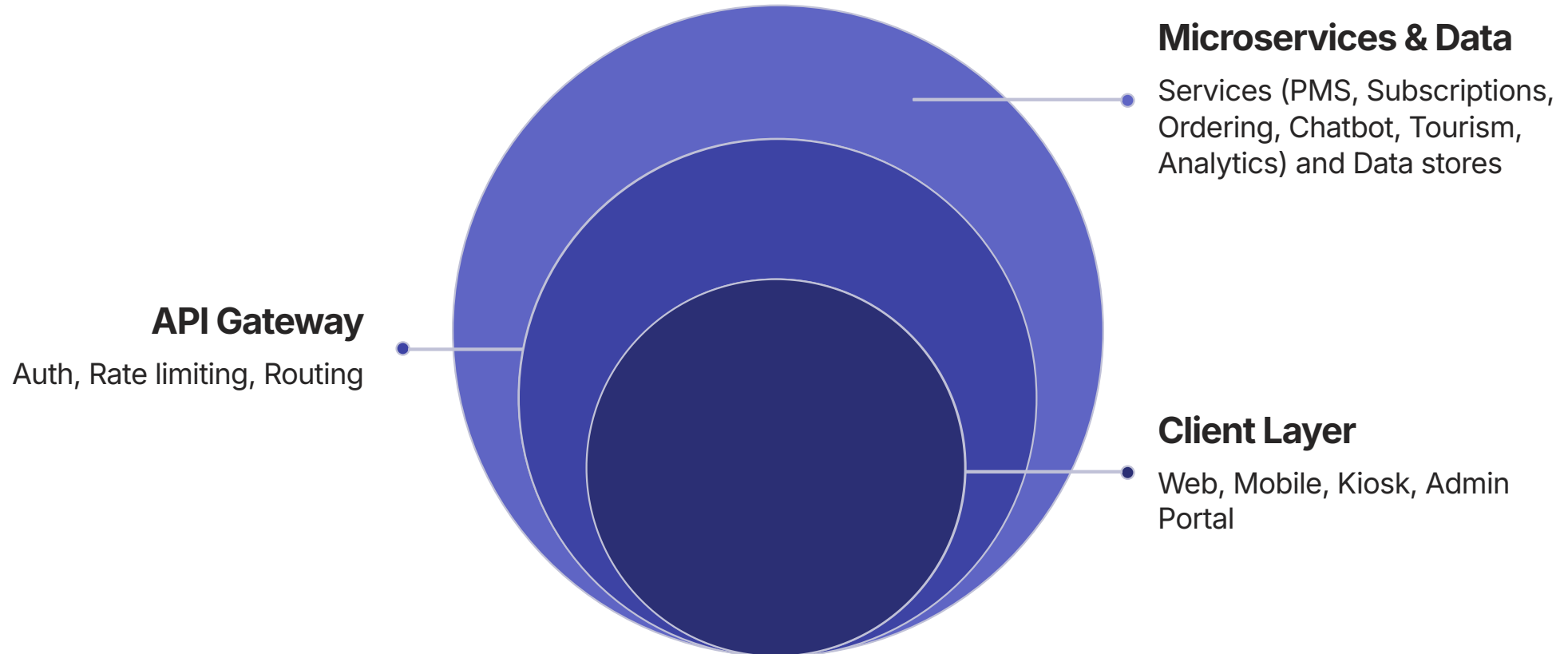
The Hotel Management Platform is organized into six interconnected domains, each serving a distinct operational or guest-facing function. The architecture follows a microservices-first approach with a unified API gateway, shared authentication layer, and centralized data lake for analytics. Each domain can be deployed independently or as part of a full-suite installation.



The platform supports multi-tenant SaaS deployments for hotel chains as well as single-property on-premise installations. Subscription tiers determine which modules are activated, with installation keys generated and validated through the licensing service. All modules communicate through a RESTful API gateway with WebSocket support for real-time features like chat and order tracking.

System Architecture & Technology Stack

The platform is built on a cloud-native, containerized architecture designed for horizontal scalability, high availability, and fault isolation. Each service is independently deployable via Kubernetes, with inter-service communication handled through gRPC for internal calls and REST/GraphQL for external APIs. The architecture supports both SaaS multi-tenant and on-premise single-tenant deployments.



The API Gateway acts as the single entry point, handling JWT-based authentication, request routing, rate limiting, and protocol translation. Services communicate asynchronously via Apache Kafka for event-driven workflows (order confirmations, subscription renewals, chatbot escalations) and synchronously via gRPC for low-latency operations like room availability checks and payment processing.

Backend Stack

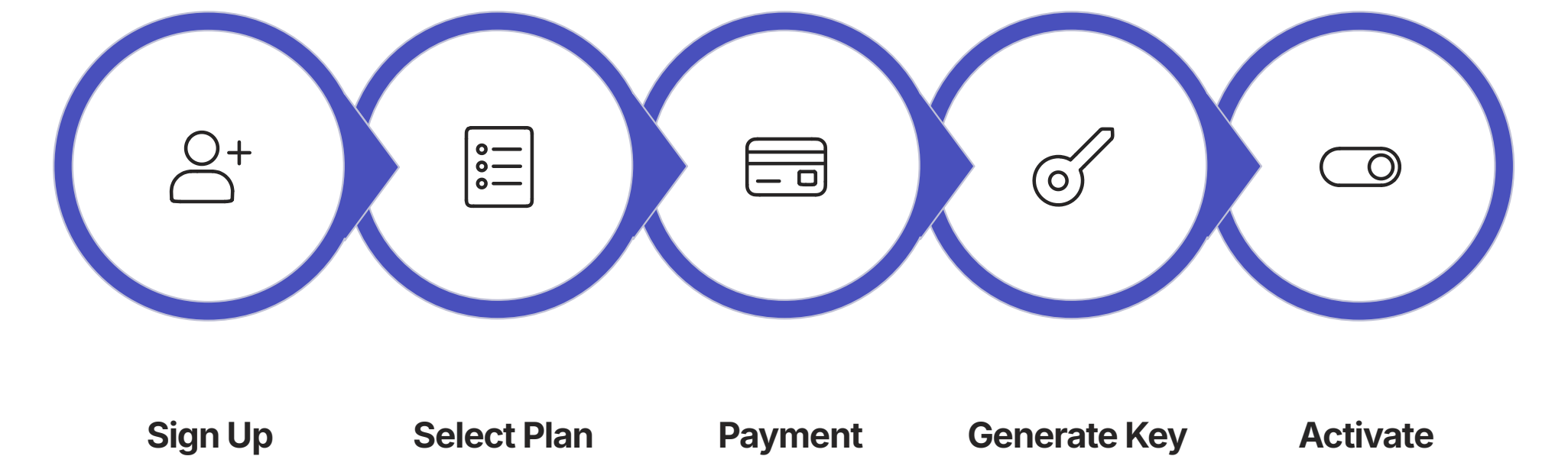
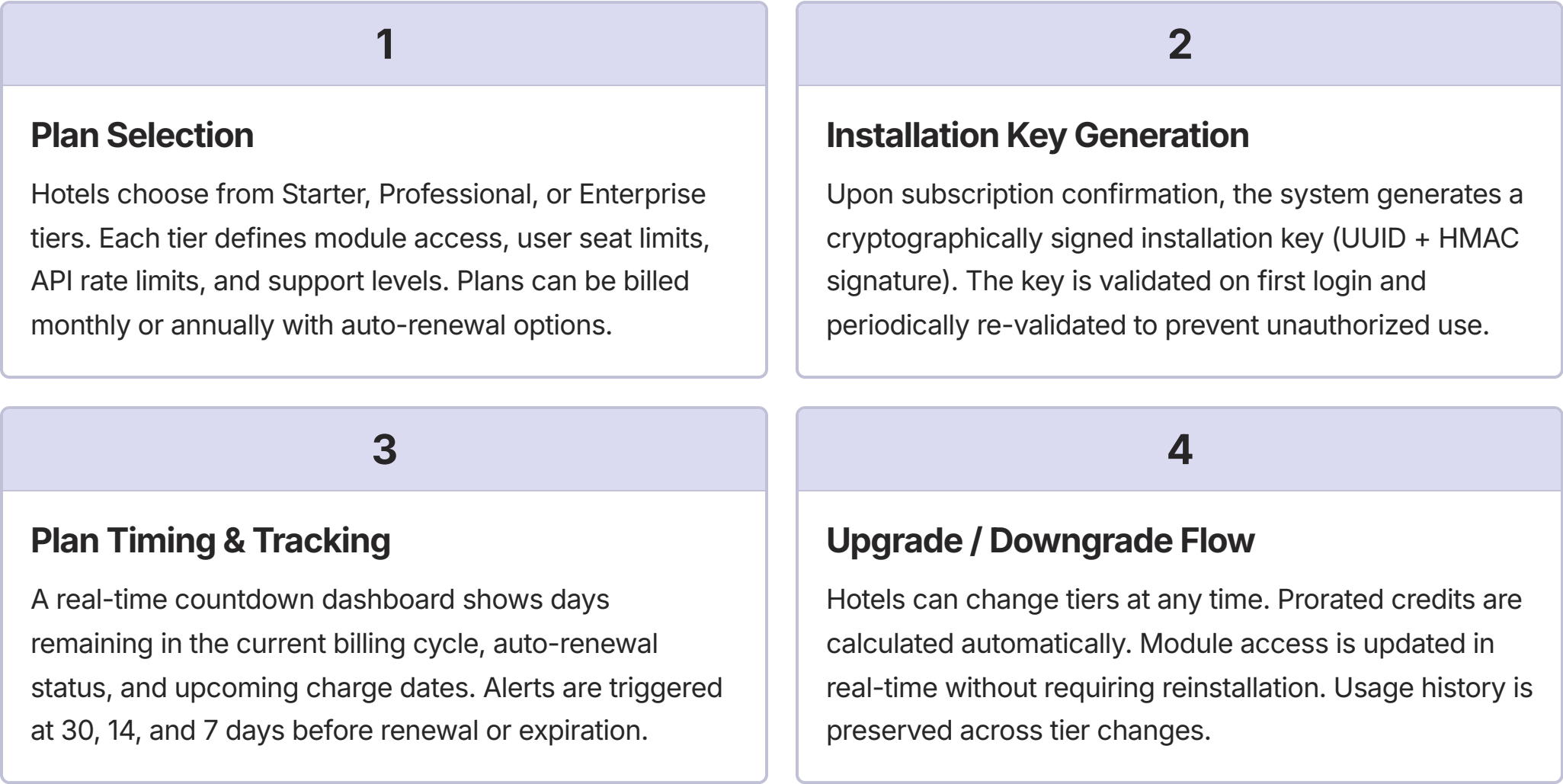
- Node.js / Python (FastAPI) for microservices
- PostgreSQL (primary DB) + Redis (caching)
- Apache Kafka (event streaming)
- Elasticsearch (search & analytics)
- gRPC + REST + GraphQL APIs

Frontend & Infrastructure

- React / Next.js (web), React Native (mobile)
- Kubernetes (container orchestration)
- AWS / Azure / GCP (cloud providers)
- Docker (containerization)
- Terraform (infrastructure as code)

Subscription & Licensing Module

The Subscription Module manages the entire commercial lifecycle of the platform — from plan selection and installation key generation to usage tracking, renewals, and upgrades. Each hotel property receives a unique installation key tied to its subscription tier, which unlocks specific modules and enforces feature gates at the API level.



The subscription service exposes webhooks for third-party billing integrations (Stripe, PayPal) and maintains an audit log of all plan changes, key activations, and payment events for compliance and dispute resolution.

Guest Ordering Module — Amenities & Supplies

The Guest Ordering Module enables hotel guests to request items directly from their room via the mobile app or in-room tablet. Orders are routed to the appropriate hotel department (housekeeping, kitchen, front desk) and tracked in real-time until delivery. The module supports both complimentary and paid items with integrated billing.

Orderable Categories



Bedding

Pillows, mattresses,
blankets, bed sheets



Beverages

Bottled water, soft
drinks, coffee, tea



Amenities

Toiletries, towels,
slippers, robes



Food

Room service, snacks,
mini-bar restock

Order Processing Workflow



Guest Places Order

Via app or in-room tablet with item selection and delivery time preference



Department Notification

Real-time push notification sent to relevant staff with order details



Fulfillment & Delivery

Staff marks order as prepared, then delivered. Guest confirms receipt in-app

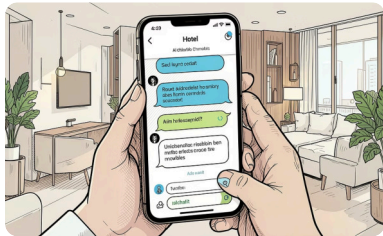


Billing & Feedback

Charges posted to room folio automatically. Guest rates the experience

AI Chatbot & Human Support Module

The platform includes a two-tier support system: an AI-powered chatbot for instant, 24/7 responses to common guest and staff queries, and an optional paid Human Support escalation path for complex issues requiring human expertise. The chatbot is trained on hotel-specific knowledge bases and integrates with all platform modules to perform actions (e.g., placing orders, checking availability).



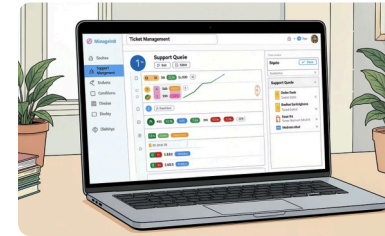
AI Chatbot Capabilities

- Natural language understanding for guest queries
- Multi-language support (20+ languages)
- Integration with PMS for room-specific context
- Can place orders, book services, answer FAQs
- Escalation trigger when confidence score drops below threshold



Human Support (Paid Tier)

- Live chat with trained hotel support specialists
- Video call option for complex technical issues
- SLA-backed response times (15 min for Enterprise)
- Screen sharing and remote assistance capability
- Full session transcript saved to ticket history



Ticketing & Escalation

- Automatic ticket creation on chatbot escalation
- Priority routing based on subscription tier
- Internal notes and assignment to specialists
- Guest-visible status updates and resolution tracking
- CSAT survey triggered on ticket closure

Tourism & Discovery Module — "Near Me"

The Tourism Discovery Module helps guests explore their surroundings by surfacing nearby points of interest, restaurants, attractions, and local experiences. Powered by integrated map APIs (Google Maps, Mapbox) and a curated content database, the module provides personalized recommendations based on guest preferences, time of day, and proximity to the hotel.

Core Features

Interactive Map View

Live map with categorized pins for restaurants, attractions, transport, and shopping within configurable radius

Curated Recommendations

Hotel-curated "Staff Picks" and AI-generated suggestions based on guest profile and past behavior

Booking Integration

Direct booking links for tours, tickets, and reservations through partner APIs (Viator, GetYourGuide, OpenTable)

Data Sources & APIs

Google Places API

POI data, ratings, photos, and real-time popularity

Mapbox GL

Custom map styling and offline map caching

TripAdvisor API

Reviews and rankings for attractions and restaurants

Hotel CMS

Property-specific curated content and promotions

Additional Core Hotel Modules

Beyond the guest-facing features, the platform includes essential operational modules that form the backbone of hotel management. These modules handle day-to-day operations from check-in to check-out, staff coordination, inventory management, and financial reporting. They are available across all subscription tiers with varying depth of functionality.



Property Management (PMS)

Room inventory, reservations, check-in/check-out, room assignments, and guest profiles. Supports walk-ins, group bookings, and OTA integrations.



Staff Management

Shift scheduling, task assignments, internal messaging, and performance tracking. Includes role-based access control for front desk, housekeeping, and management.



Billing & Invoicing

Automated folio generation, multi-payment method support, tax calculation, and integration with accounting software (QuickBooks, Xero).



Analytics & Reporting

Real-time dashboards for occupancy rates, RevPAR, ADR, guest satisfaction scores, and operational KPIs. Exportable reports in PDF and Excel.



Inventory Management

Track hotel supplies, minibar stock, linen inventory, and amenity levels. Automated low-stock alerts and purchase order generation.

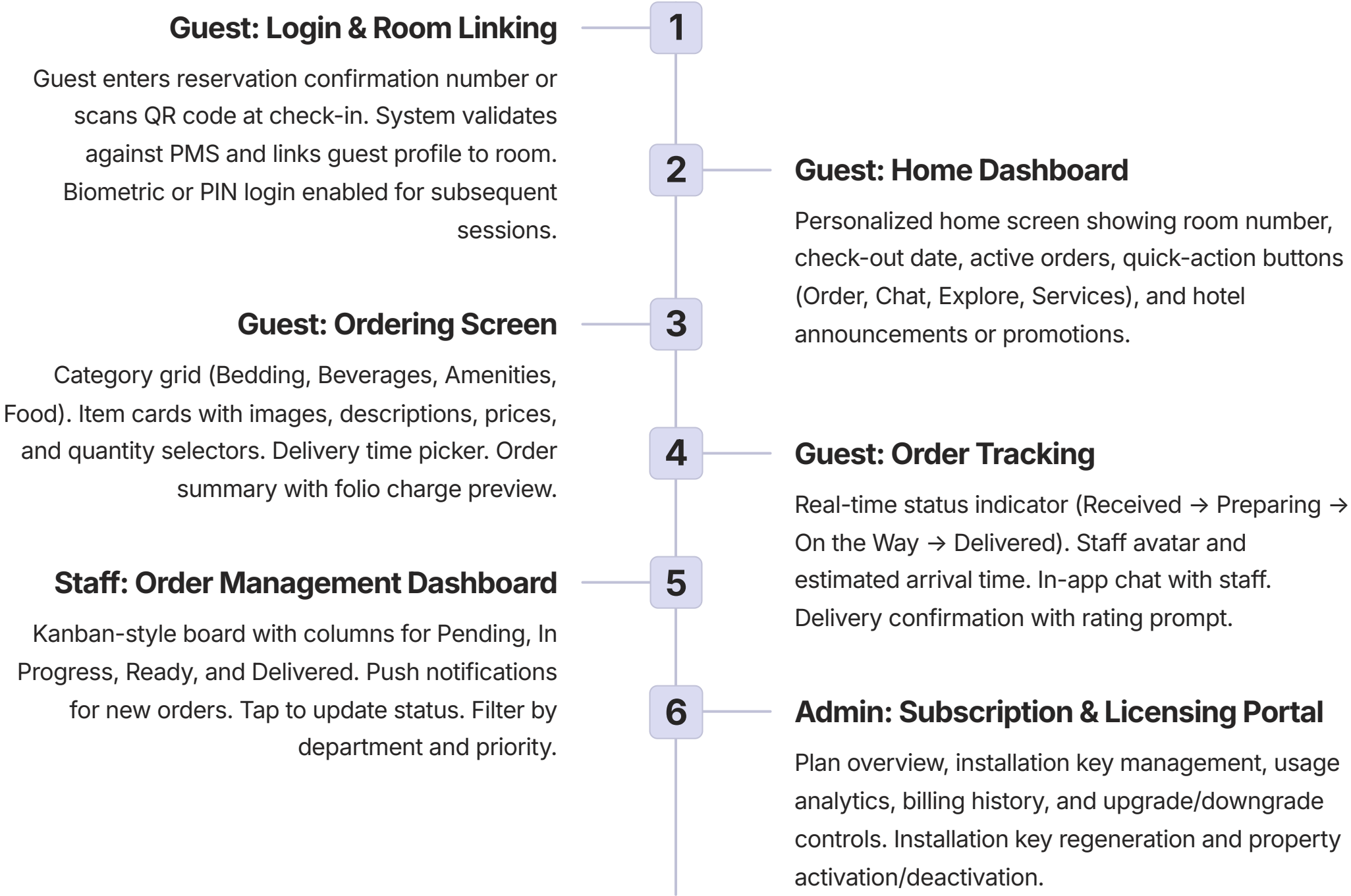


Guest Feedback & Reviews

Post-stay surveys, in-app feedback collection, review management, and sentiment analysis. Integrates with TripAdvisor and Google Reviews.

Screen-by-Screen Explanation — Key User Flows

This section provides a detailed walkthrough of the primary screens and user flows across the platform. Each screen is designed with a mobile-first approach and adapts to tablet and desktop layouts. The following covers the most critical flows for guests, hotel staff, and administrators.



Implementation Roadmap & Next Steps

The platform should be developed and deployed in phased releases to manage complexity, validate assumptions with real users, and ensure each module is production-ready before expanding scope. The following roadmap outlines a recommended 12-month development and rollout plan.



Phase 1 — Foundation (Months 1–3)

Core PMS, authentication, API gateway, database architecture, and basic staff dashboard. Establish CI/CD pipelines, Kubernetes cluster, and monitoring infrastructure.



Phase 3 — AI & Support (Months 7–9)

AI chatbot integration, human support escalation, ticketing system, and tourism discovery module. Train chatbot on hotel-specific data. Launch paid human support tier.

Key Success Metrics

- 99.9% uptime SLA across all modules
- Guest app adoption rate >60% within 30 days
- Order fulfillment time <15 minutes average
- Chatbot resolution rate >75% without escalation
- Subscription renewal rate >85%



Phase 2 — Subscription & Ordering (Months 4–6)

Subscription module with installation key generation, plan timing tracking, guest ordering module, and staff order management dashboard. Pilot with 2–3 partner hotels.



Phase 4 — Scale & Optimize (Months 10–12)

Full analytics dashboard, inventory management, guest feedback module, and multi-property support. Performance optimization, security audit, and public launch.

Recommended Team Structure

- 1 Product Manager (platform-wide)
- 2 Backend Engineers (microservices)
- 2 Frontend Engineers (web + mobile)
- 1 DevOps Engineer (infrastructure)
- 1 QA Engineer (automation + manual)
- 1 UX Designer (research + UI)

i This document serves as the foundational product specification. Each module should have its own detailed technical design document (TDD) created during the sprint planning phase, covering API contracts, database schemas, and acceptance criteria.